

Project Initialization and Planning Phase

Date	10/07/2026
Name	Karuru Arjun
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report:

The proposed solution aims to develop an AI-powered Voice-Based Concept Understanding Analyser (VBCUA) that evaluates a user's conceptual understanding and speech fluency through voice-based explanations. The system uses advanced speech recognition, semantic similarity analysis, and audio signal processing to generate comprehensive performance evaluations. This solution reduces manual assessment effort, provides objective and consistent feedback, and helps students, educators, and professionals improve both conceptual knowledge and communication skills.

Project Overview	
Objective	To develop an AI-powered application that evaluates conceptual understanding and speech fluency by analyzing users' spoken explanations using speech recognition, semantic similarity models, and audio feature extraction.
Scope	The project focuses on audio recording/upload, speech-to-text conversion, semantic similarity analysis, speech fluency evaluation, AI-based scoring, waveform visualization, dashboard reporting, and downloadable PDF report generation.
Problem Statement	
Description	Traditional evaluation of conceptual understanding is manual, time-consuming, and often subjective. Assessing both conceptual knowledge and communication skills consistently requires significant effort from educators and trainers.
Impact	An automated AI-based evaluation system improves assessment accuracy, reduces evaluation time, provides consistent feedback, and enables continuous learning through personalized performance analysis.
Proposed Solution	

Approach	Develop a Streamlit-based AI application that accepts voice recordings, converts speech into text using OpenAI Whisper, compares explanations with reference concepts using Sentence-BERT, analyzes speech fluency using Librosa, and generates detailed evaluation reports with scores and feedback.
Key Features	Voice recording and audio upload support. • Speech-to-Text conversion using OpenAI Whisper. • Semantic similarity analysis using Sentence-BERT. • Speech fluency analysis (pause ratio, RMS energy, duration, filler words). • AI-generated conceptual understanding score and qualitative feedback. • Interactive Streamlit dashboard with waveform visualization. • Downloadable PDF performance reports. • User-friendly interface for students, educators, and researchers.

Resource Requirements:

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i5/i7 Processor (or equivalent)
Memory	RAM	8 GB RAM (Minimum)
Storage	Disk space for dataset, trained model, and application	20 GB Free Storage (SSD Preferred)
Software		
Framework	Audio Processing Libraries	Librosa, SoundFile, NumPy
Libraries	Machine Learning & Data Processing	OpenAI Whisper, Sentence-Transformers, Scikit-learn
Development Environment	IDE	Visual Studio Code / PyCharm / Jupyter Notebook
Data		
Dataset	Source, size, format	Reference Concept Knowledge Base and User Audio Files (WAV/MP3/M4A)